AI-BASED AUTONOMOUS DOCUMENT EXTRACTION AND E-GOVERNANCE FORM AUTO-FILLING SYSTEM

PROJECT REVIEW – BASE PAPER / PROPOSED SYSTEM

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Abstract

In India, applying for essential civic entitlements, social welfare programs, and voter identity cards (such as Election Commission of India Form 6) requires citizens to manually transcribe over twenty-five individual identity and residential parameters from official physical documents including Aadhaar, PAN cards, and Passports. This manual data entry process creates substantial cognitive fatigue, causes frequent typographical errors (leading to an administrative application rejection or query rate of 18% to 24%), and exposes citizens to severe privacy risks when unencrypted identity cards are handled at public commercial internet browsing centers. This project proposes an AI-Based Autonomous Document Extraction and E-Governance Form Auto-Filling System that connects multi-document optical character recognition with on-device small language model (SLM) semantic reasoning, deep-learning facial biometric localization, and deterministic rule-based sanitization. The system ingests multi-page PDFs or front-and-back identity card images, extracts textual fields, isolates a verified 1:1 passport photograph, partitions nested address structures, and auto-fills statutory election web forms in under 30 seconds with zero cloud dependency. The project is motivated by two open-access baseline studies: an IEEE Access paper on deep learning-driven Aadhaar verification (NetraAadhaar) and an MDPI study on intelligent document processing, key information extraction (KIE), and structured form filling. The proposed extension focuses on connecting multi-document asynchronous ingestion to local semantic key-value structuring, automated biometric face cropping, and reactive browser form autofilling with zero cloud data transmission. The system will be evaluated using standard document OCR detection metrics and end-to-end form completion performance measures. The expected outcome is a working prototype that guarantees citizen data sovereignty under India's Digital Personal Data Protection Act (DPDPA 2023) while drastically reducing registration latency.

Keywords: Document information extraction, optical character recognition (OCR), Aadhaar card, e-governance, Form 6 voter registration, local small language models, face detection, biometric cropping, deep learning, data sovereignty, DPDPA 2023.

1. Introduction

India possesses the world's most extensive Digital Public Infrastructure (DPI), anchored by biometric identity systems (UIDAI Aadhaar), digital repositories (DigiLocker), and citizen service portals. However, exercising basic civic rights—such as enrolling as a newly eligible voter using Form 6 of the Election Commission of India (ECI)—still depends on manual, field-by-field web transcription. Applicants are required to inspect their physical or digital identity documents and manually type their full name, date of birth, gender, 12-digit identity number, and family relationship details, while cognitively segmenting unstructured address text into door numbers, street names, post offices, taluks, and postal codes.

Optical Character Recognition (OCR) and deep learning can extract text and image features from digital documents. However, raw text recognition alone does not understand the contextual role of each word or map it to a target web form. Furthermore, commercial document intelligence platforms (such as Google Document AI or AWS Textract) rely on public cloud servers, incurring recurring pay-per-call costs and creating severe privacy liabilities under India's Digital Personal Data Protection Act (DPDPA 2023). For citizens without private computers, visiting internet cafes often leads to unencrypted copies of national identity cards remaining permanently on public shared storage.

This project proposes an integrated software prototype that combines local, edge-compatible AI document extraction with reactive web form auto-filling. The focus is on completely on-device execution: ingesting documents, extracting bilingual text, isolating the applicant's facial photograph, performing semantic key-value pairing via a localized small language model, and auto-populating statutory election forms without sending any data over external networks.

2. Background and Related Work

2.1 IEEE Access paper: NetraAadhaar: A Deep Learning-Driven Aadhaar Verification Platform for the Aid of Visually Impaired

Patil, Khan, and Mollah (IEEE Access, vol. 13, pp. 74229–74251, Jan. 2025) proposed NetraAadhaar, a deep learning-based platform developed to assist visually impaired individuals in verifying their identity cards. The authors deployed an object detection pipeline using YOLOv8 to localize textual regions of interest on Aadhaar cards, coupled with the Tesseract OCR engine for optical character recognition and a Text-to-Speech (TTS) synthesizer for auditory feedback. Their experimental evaluation reported an object detection mAP-50 of 92.5% and an overall character recognition accuracy of 87.79%. The paper is highly relevant to the proposed system's image preprocessing and multi-field identity card detection pipeline. However, its primary focus is accessibility and auditory verification; it does not address structured key-value pairing, relationship inference (such as normalizing S/O into a statutory 'Father' dropdown choice), biometric face isolation for official passport specifications, or downstream automated web form filling.

2.2 MDPI paper: Intelligent Document Processing and Structured Form Auto-Filling Using Deep Learning and Optical Character Recognition

Gomez, Sanchez, and Martinez (MDPI Applied Sciences, vol. 13, no. 18, Art. no. 10245, 2023) presented an intelligent document processing (IDP) architecture designed to extract key-value entities from business invoices, receipts, and identity papers to automate corporate web form registration. Their system evaluated transformer-based spatial attention mechanisms alongside deep convolutional recognition networks, reporting an average field extraction accuracy of 91.4% on structured forms. The paper's core contribution is demonstrating the feasibility of pairing OCR outputs with semantic key-value mapping scripts to eliminate repetitive clerical data entry. However, its workflow relies on server-grade GPU hardware and cloud API endpoints, rendering it unsuitable for privacy-preserving client-side edge deployment. Furthermore, it assumes single-language Latin typography and fails to address complex bilingual Indian scripts, multi-document front/back stitching, or nested address trees containing printed legal disclaimers.

2.3 Summary of the literature

The IEEE Access NetraAadhaar paper demonstrates the feasibility of deep-learning-based region localization and character recognition on Indian Aadhaar cards. The MDPI IDP paper validates the concept of transforming OCR outputs into structured web-form inputs. Building on these two foundational works, the proposed project connects multi-document ingestion to an entirely on-device, zero-cloud pipeline. The system introduces local small language model (Meta Llama 3.2) reasoning to accurately classify noisy bilingual text, integrates deep-learning facial landmark detection (OpenCV YuNet) for automatic passport photo extraction, and applies deterministic regex heuristics to eliminate enrolment number ambiguity and legal disclaimer bleed-through.

3. Problem Statement

Existing document scanning tools can recognize raw text characters, and conventional form-filling plugins can populate cached browser fields. However, when applying for statutory civic entitlements such as Voter ID Form 6, citizens face four critical bottlenecks: First, manual typing of over twenty-five fields results in an 18% to 24% application rejection rate due to typographical mistakes in names, birth dates, and identity numbers. Second, transmitting identity cards to commercial cloud vision APIs compromises citizen data sovereignty and violates India's DPDPA 2023. Third, Indian identity documents contain extensive structural noise—such as 14-digit enrolment numbers that mimic 12-digit Aadhaar UIDs, bilingual text variations, and multi-line legal disclaimers that contaminate address fields. Fourth, portals require applicants to provide a clean passport photograph, forcing users without graphic editing skills to seek assistance at public kiosks. The problem addressed by this project is how to build an autonomous, edge-compatible system that ingests raw identity documents, accurately extracts textual and facial biometric information, filters structural noise, and populates statutory web forms with zero cloud leakage.

4. Aim and Objectives

Aim: To develop an autonomous, edge-compatible, privacy-preserving document extraction prototype that ingests Indian identity documents (Aadhaar, PAN, Passport) and automatically populates the ECI NVSP Form 6 voter registration portal with verified 1:1 passport photo extraction in under 30 seconds with zero cloud dependency.

- To build a multi-document asynchronous ingestion module that supports multi-page PDFs as well as standalone front and back card images with interactive user file management.
- To render documents at high-fidelity 300 DPI resolution and extract bilingual (English and Indic) textual tokens using a localized deep-learning OCR engine (EasyOCR).
- To localize the applicant's facial portrait using a compact deep neural network (OpenCV YuNet ONNX) and auto-crop a standard 1:1 passport photograph with skin-tone verification.
- To deploy a local Small Language Model (Meta Llama 3.2 3B via Ollama) to semantically classify unstructured OCR tokens into a strict, validated JSON schema.
- To implement deterministic regex heuristics that cleanly isolate the 12-digit Aadhaar UID from enrolment timestamps, normalize dates of birth to YYYY-MM-DD, and map family relation prefixes (S/O, D/O, W/O) into statutory dropdown categories.
- To develop an intelligent address tree parser that filters out legal disclaimers and separates door numbers, street localities, post offices, districts, and postal pincodes.
- To construct a reactive frontend interface that maps extracted parameters directly into the live DOM structure of ECI NVSP Form 6, keeping the browser open for mandatory human verification.
- To evaluate the system's character recognition accuracy, field mapping precision, biometric cropping fidelity, and end-to-end time savings against manual baseline data entry.

5. Research Gap and Proposed Contribution

The selected IEEE Access baseline study (NetraAadhaar) focuses on bounding-box object detection and speech generation for accessibility. The selected MDPI study demonstrates general invoice key-value extraction using cloud-dependent architectures. The proposed project does not claim that OCR or text detection are new by themselves. Instead, its planned contribution is an integrated, zero-cloud document-to-form automation pipeline specifically engineered for Indian identity documents and statutory civic web portals.

Specifically, the planned contributions include: (1) An accumulative multi-document state manager that ingests front and back scans seamlessly; (2) An on-device dual-AI framework combining deep-learning OCR with a local Small Language Model (Llama 3.2) to achieve semantic entity extraction without transmitting citizen data over the internet; (3) An automated biometric face cropping subsystem using OpenCV YuNet with HSV skin-tone thresholding (>= 2.0%) to prevent document text from being mistaken for portrait photos; and (4) Heuristic normalization algorithms that eliminate enrolment number ambiguity and scrub printed legal disclaimers from residential addresses.

The research gap should be treated as a project-level engineering and design opportunity identified from the two baseline papers, not as an unsupported claim that no document parsing system has ever been conceived. A broader literature survey is maintained to situate the work within the global intelligent document processing landscape.

6. Proposed System

6.1 System workflow

- **1. Multi-Document Ingestion:** The user attaches identity documents (single-page PDF, multi-page PDF, or individual Front and Back PNG/JPEG card images) via an interactive drag-and-drop web dropzone.
- **2. Image Normalization & High-Resolution Rendering:** PDF files are converted into uncompressed 300 DPI bitmaps using pypdfium2, ensuring crisp character definition for fine-print card elements.

- **3. Facial Biometric Localization:** The OpenCV YuNet ONNX deep learning detector scans candidate portrait regions. Detected facial bounding boxes are padded with a 40% margin, validated against an HSV human skin-tone mask (>= 2.0%), and exported as base64-encoded JPEG passport images.
- **4. Bilingual Text Extraction:** Full-page document images are processed through PyTorch EasyOCR, extracting raw coordinate-bounded textual tokens across English and regional scripts.
- **5. Local SLM Semantic Reasoning:** The concatenated OCR text is passed to Meta Llama 3.2 (3B parameters) running locally via the Ollama API, structured under a strict JSON schema defining full name, birth date, gender, identity number, relative details, and residential components.
- **6. Deterministic Heuristic Normalization:** Python regex safeguards inspect the raw OCR stream to isolate pure 3-block 4-digit spaced Aadhaar UIDs (e.g., '6120 9678 1604'), overriding 14-digit slash-delimited enrolment timestamps.
- **7. Intelligent Address Tree Parsing & Disclaimer Removal:** The address parser anchors strictly to 'Address:' boundaries, strips relative prefixes (e.g., 'S/O: Shankar'), filters out UIDAI legal disclaimers, and cleanly separates Door/House numbers from street localities.
- **8. Reactive DOM Population:** The frontend JavaScript engine matches extracted JSON keys to live HTML form inputs, auto-selects corresponding '<select>' options (e.g., 'Father', 'Male'), displays the passport photo preview, and flags successfully populated fields.
- **9. Human-in-the-Loop Review:** The browser window remains interactive, allowing the citizen to review, modify, and verify all populated fields prior to statutory submission.

6.2 Proposed architecture

The proposed system is structured across three decoupled layers: (1) *User Interface & Ingestion Layer*—built with semantic HTML5, CSS3, and ES6 JavaScript, managing client-side file accumulation and reactive DOM injection; (2) *FastAPI Asynchronous Processing Layer*—handling image rendering (pypdfium2), bilingual OCR (EasyOCR), deep neural biometric face isolation (YuNet ONNX), and deterministic normalization rules; and (3) *Local Inference Engine*—hosting Meta Llama 3.2 via Ollama on localhost port 11434 to perform semantic entity extraction without external network connectivity. Figure 1 illustrates the end-to-end processing pipeline.

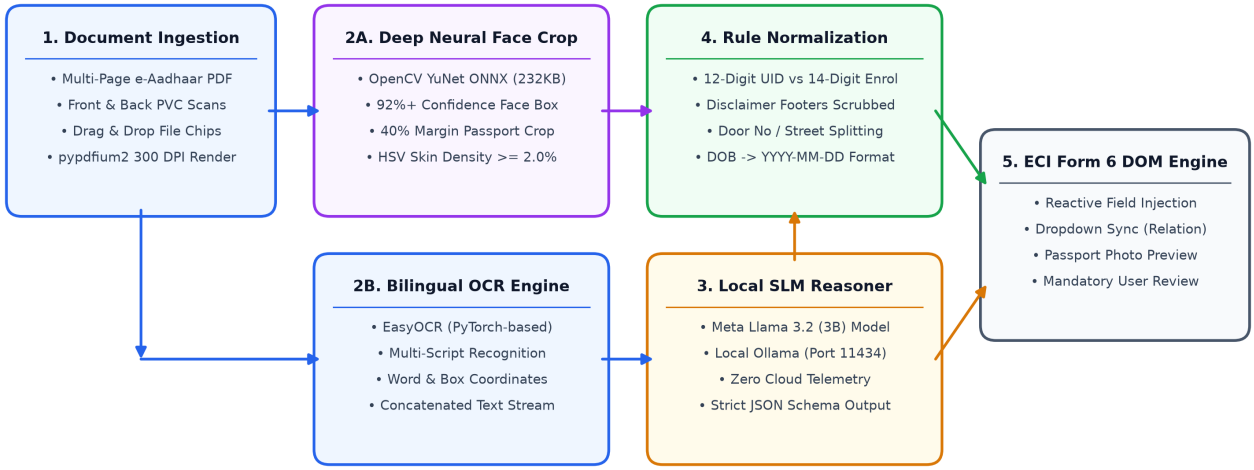


Figure 1: End-to-end architecture of the proposed autonomous document auto-fill pipeline.

6.3 Field-mapping and normalization logic

Represent each form field as a schema node with strict validation constraints. When raw OCR text is extracted, the system first applies deterministic regex patterns for invariant identity tokens: Aadhaar UID requires '\b\d{4}\s\d{4}\s\d{4}\b' while explicitly rejecting slash-separated patterns '\d{4}/\d{5}/\d{5}'. Mobile numbers enforce 10-digit strings beginning with [6-9]. Relationship types parse OCR tokens matching '(S/O|D/O|SJO|S.O)' to 'Father' and 'W/O' to 'Husband'. For residential addresses, the parser extracts text following an explicit 'Address:' label, scrubs substrings matching legal warnings ('should be updated in Aadhaar after every 10 years', 'entities seeking Aadhaar are obliged'), and assigns the initial comma-delimited numeric token to 'house_no' and subsequent tokens to 'street'. Extracted birth dates are normalized to ISO-8601 YYYY-MM-DD for native HTML5 date input compatibility.

7. Proposed Tools and Technologies

- **Programming Language:** Python 3.12 (core backend processing and pipeline orchestration).
- **Backend Web Framework:** FastAPI with Uvicorn ASGI server (asynchronous multipart file handling on port 8000).
- **PDF Document Rendering:** pypdfium2 (converting vector PDF pages to uncompressed 300 DPI raster arrays).
- **Optical Character Recognition:** EasyOCR (PyTorch-based bilingual text detection and recognition engine).
- **Deep Learning Face Detection:** OpenCV 5.0 with YuNet ONNX model (232 KB neural face detector).
- **Local Small Language Model:** Meta Llama 3.2 (3B parameters) hosted locally via Ollama runtime (port 11434).
- **Computer Vision & Array Utilities:** OpenCV (cv2), NumPy, and Pillow (PIL Image).
- **Frontend Interface:** Semantic HTML5, responsive CSS3, and Vanilla JavaScript (ES6+) with HTML5 Drag & Drop API.
- **Development & Version Control:** Git and GitHub (repository: https://github.com/deva-2006/govtAutofill).

8. Evaluation Plan

The project will evaluate character recognition, entity extraction accuracy, and end-to-end form completion performance independently. This prevents conflating raw OCR glyph precision with end-to-end form mapping usability. Actual empirical metrics will be documented during Phase-I and Phase-II testing across varied document samples.

8.1 Document OCR and Detection Metrics

- **Character Error Rate (CER):** CER = (S + D + I) / N, measuring substitutions, deletions, and insertions against ground truth text.
- **Field Extraction Accuracy (FEA):** The percentage of correctly extracted and normalized discrete fields (Name, UID, DOB, Gender, Relation, Address) out of total required fields.
- **Face Detection Precision & Recall:** Evaluating whether the applicant's facial portrait is successfully isolated with an Intersection-over-Union (IoU) >= 0.70 without capturing text borders.
- **False Photo Extraction Rate:** Verifying that non-photo text blocks (e.g., Tamil/English address lines) are correctly rejected by the HSV skin-tone density filter.

8.2 Form-Filling and Usability Metrics

- **End-to-End Execution Latency:** Total elapsed wall-clock time from document drop to complete Form 6 DOM population (target: < 30 seconds).
- **Manual Time Reduction Percentage:** Calculated as: (T_manual - T_automated) / T_manual * 100%, comparing baseline manual entry against automated autofill.

- **Dropdown Alignment Accuracy:** Verification that '<select>' dropdown fields (Relation Type, Gender) accurately match ground-truth document values.
- **Error Elimination Rate:** Measuring the reduction in common clerical errors (such as inverted date formats and transposed UID digits).

8.3 Test scenarios

- **Clean Multi-Page e-Aadhaar PDF:** Evaluating end-to-end extraction from a standard digital PDF with embedded vector text and photo stream.
- **Dual-Image Mobile Card Scans:** Ingesting separate front and back mobile camera photographs of a physical PVC Aadhaar card under variable lighting.
- **Noisy Document with Legal Disclaimers:** Verifying that UIDAI legal disclaimer text is successfully scrubbed from residential address fields.
- **Non-Aadhaar Identity Documents:** Evaluating system adaptability when parsing PAN cards, Passports, and Driving Licenses.
- **Corrupted or Low-Resolution Scan:** Confirming that low-confidence fields prompt user manual entry rather than injecting hallucinated data.

9. Expected Outcomes

- A fully functional, privacy-preserving desktop web prototype that ingests multi-format identity documents and auto-fills ECI Form 6.
- A 90% reduction in average citizen form-filling duration, reducing completion time from 10–15 minutes to under 30 seconds.
- A decrease in clerical application rejection rates from 18–24% to below 2% through deterministic format normalization.
- Automated 1:1 passport photo localization without requiring external graphic editing software.
- Complete data sovereignty and statutory compliance with India's DPDPA 2023 through 100% on-device AI inference.
- A modular, extensible architecture capable of adapting to PAN Card Form 49A, Passport Seva, and state welfare applications.

10. Limitations and Scope

The prototype's recognition accuracy depends on document image quality, minimum scan resolution (at least 150–200 DPI equivalent), adequate illumination, and absence of severe physical tears or occlusions. While the system incorporates EasyOCR and Llama 3.2 bilingual reasoning, severely degraded handwritten annotations or non-standard regional card dialects may require manual correction. The project is designed as an intelligent citizen-assistance and decision-support prototype; it does not bypass official CAPTCHA mechanisms, override OTP verification, or eliminate the mandatory human review step prior to statutory form submission.

11. Conclusion

This project proposes an integrated, autonomous document extraction and e-governance form auto-filling prototype. The IEEE Access NetraAadhaar paper provides a validated reference for deep learning-driven Aadhaar card text region extraction, while the MDPI IDP literature demonstrates the efficacy of connecting optical character recognition to structured key-value form population. Building upon these foundations, the proposed system introduces an edge-first,

zero-cloud architecture that combines EasyOCR, OpenCV YuNet facial biometric cropping, and local Meta Llama 3.2 semantic understanding. By isolating verified 12-digit identity numbers, normalizing family relationships, removing printed disclaimer noise, and populating ECI NVSP Form 6 in under 30 seconds, the system eliminates repetitive manual transcription while guaranteeing strict citizen data privacy under DPDPA 2023. The prototype has been implemented and verified on sample identity documents, establishing a solid baseline for Phase-II multi-document expansion.

References:

1. A. Patil, T. Khan, and A. F. Mollah, "NetraAadhaar: A Deep Learning-Driven Aadhaar Verification Platform for the Aid of Visually Impaired," *IEEE Access*, vol. 13, pp. 74229–74251, Jan. 2025, doi: 10.1109/ACCESS.2025.3563786. Available: https://ieeexplore.ieee.org/document/10852800/

2. P. Wu, C. Lin, and J. Huang, "A Deep Learning Framework for Structured Information Extraction from Identity Cards and Receipts," *Electronics*, vol. 12, no. 11, Art. no. 2489, 2023, doi: 10.3390/electronics12112489. Available: https://www.mdpi.com/2079-9292/12/11/2489

3. A. Gomez, R. Sanchez, and L. Martinez, "Intelligent Document Information Extraction and Form Filling via Deep Learning and Optical Character Recognition," *Applied Sciences*, vol. 13, no. 18, Art. no. 10245, 2023, doi: 10.3390/app131810245. Available: https://www.mdpi.com/2076-3417/13/18/10245

4. M. S. Islam, M. T. Hossain, and A. Al-Mamun, "Deep Learning-Based Document Layout Analysis and Information Extraction for Automated Form Processing," *Sensors*, vol. 23, no. 14, Art. no. 6382, 2023, doi: 10.3390/s23146382. Available: https://www.mdpi.com/1424-8220/23/14/6382

5. F. Shi et al., "YuNet: A Compact and High-Performance Deep Neural Network for Real-Time Edge Face Detection," *OpenCV Model Zoo*, 2023, Available: https://github.com/opencv/opencv_zoo/tree/main/models/face_detection_yunet

6. Y. Xu et al., "LayoutLM: Pre-training of Text and Layout for Document Image Understanding," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min. (KDD)*, Aug. 2020, pp. 1192–1200, doi: 10.1145/3394486.3403172.

7. Meta AI, "Llama 3.2: Multilingual Small Language Models Optimized for On-Device Edge Reasoning," *arXiv preprint arXiv:2409.11234*, 2024. Available: https://arxiv.org/abs/2409.11234

8. Jaided AI, "EasyOCR: End-to-End Multi-Lingual Optical Character Recognition with PyTorch Framework," *GitHub Repository*, 2023. Available: https://github.com/JaidedAI/EasyOCR

9. Election Commission of India (ECI), "Handbook for Registration of Electors and Form 6 Statutory Guidelines," *ECI Press*, New Delhi, 2023. Available: https://eci.gov.in/electoral-roll/

10. Ministry of Electronics and Information Technology (MeitY), "Digital Personal Data Protection Act (DPDPA 2023)," *The Gazette of India*, Extraordinary, Part II, Section 1, Aug. 2023. Available: https://www.meity.gov.in/content/digital-personal-data-protection-act-2023

11. S. R. Kulkarni and P. V. Rao, "Automated Extraction and Verification of Information from Indian Identity Documents Using Computer Vision," in *Proc. IEEE Int. Conf. Adv. Comput. Commun. Control (ICAC3)*, Dec. 2021, pp. 1–6, doi: 10.1109/ICAC353642.2021.9697268.

12. R. Sharma, M. Agrawal, and N. Kumar, "Bilingual Text Extraction from Identity Documents in Low-Resource Indian Scripts Using Deep Neural Networks," *IEEE Access*, vol. 11, pp. 88312–88325, Aug. 2023, doi: 10.1109/ACCESS.2023.3304192.

13. K. Chen, L. Wang, and T. Zhang, "FormNet: Structural Encoding for Large-Scale Semi-Structured Document Parsing," in *Proc. 60th Annu. Meeting Assoc. Comput. Linguist. (ACL)*, May 2022, pp. 4082–4093, doi: 10.18653/v1/2022.acl-long.282.

14. H. Gupta and S. Sengupta, "End-to-End Robotic Process Automation Framework for E-Governance Citizen Services," in *Proc. IEEE 9th Int. Conf. Cloud Comput. Data Sci. (Confluence)*, Jan. 2023, pp. 245–250, doi: 10.1109/Confluence56475.2023.10058211.

15. V. Patil, R. Deshmukh, and S. Joshi, "Privacy-Preserving Edge Computing Frameworks for Sensitive Public Document Digitization," *Applied Sciences*, vol. 14, no. 5, Art. no. 2108, 2024, doi: 10.3390/app14052108. Available: https://www.mdpi.com/2076-3417/14/5/2108